CSCE 2501 — Fundamentals of Database Systems

# Relational Model (Schema Mapping)

## Resale Real Estate Database — Milestone I

**Notation & Conventions:**
- **Underlined** attributes denote **Primary Keys (PK)**.
- ***Italicized*** attributes denote **Foreign Keys (FK)**.
- All relation schemas are fully normalized to at least **3NF / BCNF**.

## 1. Relational Schemas

### 1. Location [Independent Base Entity]

```
Location(location_id, city, district, area_name, compound_name)
```

- **PK:** location_id
- **FK:** None (Independent entity)

### 2. Developer [Independent Base Entity]

```
Developer(developer_id, developer_name, corporate_details, total_projects,
total_properties_listed)
```

- **PK:** developer_id
- **UNIQUE:** developer_name
- **FK:** None (Independent entity)

### 3. Development [Dependent Entity]

```
Development(development_id, project_name, description, developer_id, location_id)
```

- **PK:** development_id
- **FK 1:** developer_id → Developer(developer_id) ON DELETE RESTRICT ON UPDATE CASCADE
- **FK 2:** location_id → Location(location_id) ON DELETE RESTRICT ON UPDATE CASCADE

### 4. Property [Dependent Entity]

```
Property(property_id, title, description, source_url, total_area_sqm, num_bedrooms,
num_bathrooms, property_type, asking_price, price_per_sqm, payment_method,
down_payment, contract_year, discount_over_market, gain_over_market, development_id,
location_id)
```

- **PK:** property_id
- **FK 1:** development_id → Development(development_id) ON DELETE CASCADE ON UPDATE CASCADE
- **FK 2:** location_id → Location(location_id) ON DELETE RESTRICT ON UPDATE CASCADE
- **CHECK:** property_type IN ('apartment', 'villa', 'townhouse', 'duplex', 'chalet', 'penthouse', 'studio', 'other')
- **CHECK:** payment_method IN ('cash', 'installment')

### 5. Amenity [Independent Base Entity]

```
Amenity(amenity_id, amenity_name)
```

- **PK:** amenity_id
- **UNIQUE:** amenity_name
- **FK:** None (Lookup entity)

### 6. Property_Amenity [Junction Table (M:N)]

```
Property_Amenity(property_id, amenity_id)
```

- **COMPOSITE PK:** (property_id, amenity_id)
- **FK 1:** property_id → Property(property_id) ON DELETE CASCADE ON UPDATE CASCADE
- **FK 2:** amenity_id → Amenity(amenity_id) ON DELETE CASCADE ON UPDATE CASCADE

### 7. User [Independent Base Entity]

```
User(user_id, username, email, password_hash, gender, age, birthdate)
```

- **PK:** user_id
- **UNIQUE:** username, email
- **CHECK:** gender IN ('Male', 'Female')
- **CHECK:** age >= 18

### 8. User_Focus_Area [Junction Table (M:N)]

```
User_Focus_Area(user_id, location_id)
```

- **COMPOSITE PK:** (user_id, location_id)
- **FK 1:** user_id → User(user_id) ON DELETE CASCADE ON UPDATE CASCADE
- **FK 2:** location_id → Location(location_id) ON DELETE CASCADE ON UPDATE CASCADE

### 9. Review [Dependent Entity]

```
Review(review_id, user_id, development_id, rating, review_comment, review_date)
```

- **PK:** review_id
- **FK 1:** user_id → User(user_id) ON DELETE CASCADE ON UPDATE CASCADE
- **FK 2:** development_id → Development(development_id) ON DELETE CASCADE ON UPDATE CASCADE
- **CHECK:** rating BETWEEN 1 AND 10
- **UNIQUE:** (user_id, development_id) — 1 review per user per project

## 2. Normalization Verification

| Normal Form | Status | Justification |
|---|---|---|
| **1NF** | ✔ **Satisfied** | All attributes are atomic values. Multi-valued attributes (amenities, focus areas) are resolved into dedicated junction relations. |
| **2NF** | ✔ **Satisfied** | No partial dependencies exist. All non-key attributes fully depend on the complete Primary Key in every relation. |
| **3NF** | ✔ **Satisfied** | No transitive dependencies exist. Every non-key attribute depends directly and solely on the primary key. |
| **BCNF** | ✔ **Satisfied** | Every functional dependency determinant is a candidate key in all relations. |

## 3. Table Creation Order & Dependency Tree

```
Independent Base Tables (Create First):
  1. Location
  2. Developer
  3. User
  4. Amenity

Dependent & Junction Tables (Create After Parents):
  5. Development    ──> [depends on: Developer, Location]
  6. Property       ──> [depends on: Development, Location]
  7. Property_Amenity ──> [depends on: Property, Amenity]
  8. User_Focus_Area  ──> [depends on: User, Location]
  9. Review         ──> [depends on: User, Development]
```